

1 ▶ 800 CE

c 25–50 CE

Medical encyclopedia

In the early years of the Roman Empire, great advances were made in the field of medicine. At the beginning of the 1st century CE, a writer called Aulus Cornelius Celsus produced an important encyclopedia entitled *De Medicina*, which gave an up-to-date account of medicine at the time. The work included a description of surgery for kidney stones.



Later Latin edition of Soranus's work

c 100 CE

Papermaking

Around this date, true paper, as we know it today, was invented by Cai Lun, a Chinese court official. (A type of paper had already been in use for some 200 years.) Cai Lun made paper by drying out a pulp of tree bark and old rags on a screen, producing strips that could be written on.

100 CE

Health for women

Soranus, a doctor from the ancient Greek city of Ephesus, produced the first major book on women's health. He wrote about childbirth and the care of babies, including how to make feeding bottles.

127–141 CE

Ptolemy's astronomy

Greek-Roman astronomer Ptolemy devised a model to explain the movement of planets. His scheme took account of the way in which some planets appear to orbit in opposite directions from others. Ptolemy also worked out a system for measuring the latitude and longitude of places in the known world, which made it possible to create a world map.



Ptolemy's world map (14th-century version)

1

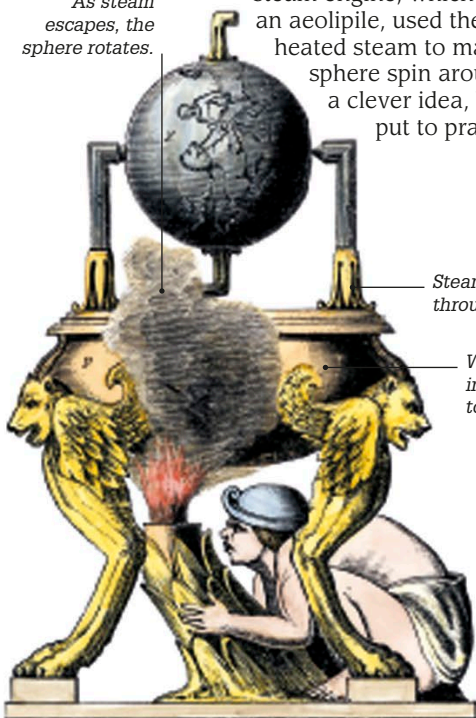
200

c 50 CE

Hero's steam engine

Greek inventor Hero devised a large number of machines. His steam engine, which he called an aeolipile, used the force of heated steam to make a metal sphere spin around. It was a clever idea, but never put to practical use.

As steam escapes, the sphere rotates.



Steam rises through tubes

Water is heated in a container to make steam.

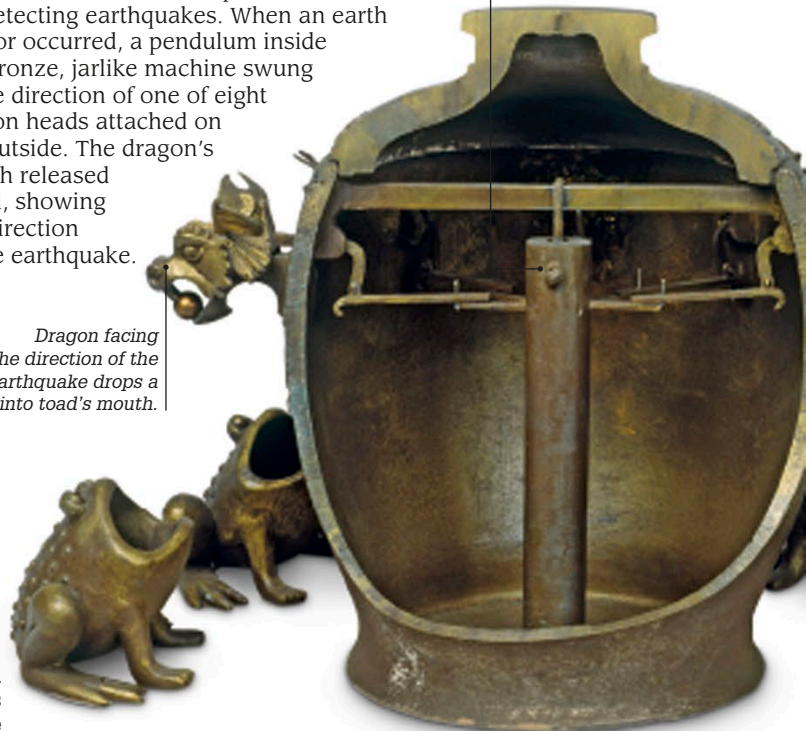
132 CE

Earliest earthquake detector

Chinese scholar Zhang Heng built the earliest-known seismoscope, an instrument for detecting earthquakes. When an earth tremor occurred, a pendulum inside the bronze, jarlike machine swung in the direction of one of eight dragon heads attached on the outside. The dragon's mouth released a ball, showing the direction of the earthquake.

Pendulum moves because of an earth tremor, operating a crank that opens the dragons' mouths.

Dragon facing the direction of the earthquake drops a ball into toad's mouth.



Cut-away model of Zhang Heng's seismoscope